

This is sample data provided for CALIBER purposes only	FIXED-BED REACTOR DATA SHEET	DOC NO: TJC-LLD-DS-DC-3401A REV: 3
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SDK Polyolefin Expansion Project

ITEM / TAG NO.	DC-3401A	EQUIPMENT ID	EQ-1000034011
EQUIPMENT NAME	CATALYST REDUCTION REACTOR	TYPE	Fixed-Bed Reactor (Pd Catalyst)
PLANT / UNIT	LINEAR LOW DENSITY POLYETHYLENE (LLDPE) UNIT	AREA	3400 - CATALYST PREPARATION
FUNCTIONAL LOC.	TJC-LLD-3400-01	CRITICALITY	HIGH CRITICAL
SERVICE	Reduction of PdO catalyst to Pd using H2 (PdO + H2 -> Pd)		

DESIGN & MECHANICAL DATA

REACTOR TYPE	VERTICAL FIXED-BED, DOWNFLOW	CATALYST	Pd / Al2O3 (PdO reduced to Pd)
ITEM NO.	DC-3401A	CATALYST VOLUME	3.2 m3
DIAMETER (ID)	1200 mm	SHELL MATERIAL	SA-516 Gr.70 + SS304 clad
TANGENT-TANGENT	4500 mm	INTERNALS	SS304 support grid + hold-down
DESIGN PRESSURE	6 barg / FV	HEATER	EA-3401 ELECTRIC PRE-HEATER 180 kW
OPERATING PRESSURE	2 kg/cm2g	INERT MEDIUM	NITROGEN
DESIGN TEMP.	250 degC	REDUCTANT	HYDROGEN (diluted in N2)
OPERATING TEMP.	180-210 degC	PSV	PSV-3401 set 6 barg

PERFORMANCE / NOZZLE DATA

EA-3401 HEATER OUTPUT	180 kW	AREA CLASSIFICATION	ZONE 1, IIC T3
HEATER VOLTAGE	380 V / 3 Ph / 50 Hz	EX PROTECTION	Ex d IIC T3
HEATER CURRENT	273 A	TIC CONTROL	TIC-17343 (heater outlet)
HEATER TYPE	IN-LINE ELECTRIC, N2 STREAM	BED TC's	TE-3401-1..8 multipoint
VENDOR / MANUFACTURER	Vendor ABC	DRIVER SUPPLIER	Supplier XYZ
DRIVER TYPE	None (static vessel) - assoc. heater EA-3401	DATASHEET REV	Rev 3 - ISSUED FOR OPERATION

NOTES:

- Cross-reference for tag DC-3401A: GA Drawing, P&ID TJC-LLD-PID-3401, Interlock SEQ-3401, OPL sheets and Maintenance History.
- Material of construction per WPN-MAT-001. Refer to interlock trip settings before operation.